What *can* can and can't do

Bronwyn M. Bjorkman and Sali A. Tagliamonte
(bronwyn.bjorkman@queensu.ca) (sali.tagliamonte@utoronto.ca)

Background

Modals form a small and well-defined class in English, united by **syntax**, **morphology**, and **semantics**

- *must*, *may*, *might*, *shall*, *should*, *will*, *would*, *can*, *could* (+ *ought*, *need*, *dare*, marginally)

- But evidence that modals are **declining** in use, being replaced by corresponding semi-modals.
  - *must* > *have to* (Tagliamonte and Smith, 2006; Tagliamonte and D'Arcy, 2007)
  - *will* > *going to* (Berglund, 1997; Szmrecsanyi, 2003; Tagliamonte, 2002)

What about *can* > *able to*?

- Google Ngrams: all modals decreasing—**except *can***:

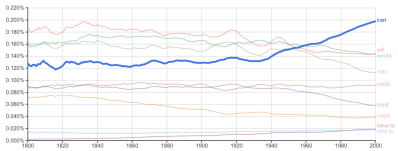


Fig. 1: Modals and semi-modals from 1800–2000 (Michel et al., 2011)

Questions:

1. Are changes in *can* reflected in **spoken** English?
2. Is there evidence of a **change in progress**?
3. Do **grammatical** factors condition the increase in *can*?

In all three cases: **Yes**.

Main Proposal:

can is diverging from the rest of the modal system because it is **structurally exceptional**, associated with a **uniquely low** position.

The corpus + study

Modals examined: all **possibility** modals (*can*, *could*, *may*, *might*) + semi-modal *able to*.

Source: York English Corpus (YEC: Tagliamonte, 1996-1998)

- Sociolinguistic interviews conducted in 1997
- Ages 15–91 (Born 1906–1982)
- Variety of social, economic, and educational backgrounds

Coding: for demographic properties of the speaker, and for a range of grammatical properties. Focus here: **modal interpretation**.

Analysis: using **GoldVarb** (Sankoff et al., 2005)

A dynamic increase in *can*

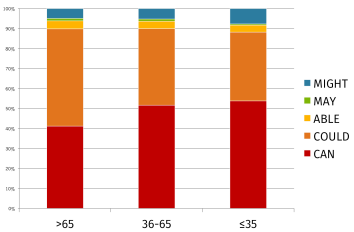


Fig. 2: Overall distribution of possibility modals by age

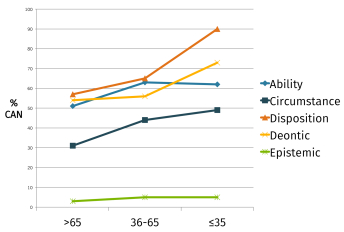


Fig. 3: % *can* for different modal interpretations

Overall increase in *can*.
Significant effects of:

age (younger speakers use more *can*)
subject type (generic and 3rd person)
interpretation (see Fig 3)

- weak effect with root meanings (ability, circumstantial, deontic)
- no increase in epistemic meanings
- sharp rise in **dispositional** / **quantificational**

questions and **negation**

Confirmation: *can* is increasing in apparent time (significant effect of age).

Dispositional interpretations

(1) a. the North-Sea [...] it can be so rough it sort-of goes along sideways. (Spkr 30; Age 20)
b. They can be quite expensive, can't they? (Spkr 64; Age 46)
c. So that can be really tedious, marking. I don't like marking. (Spkr 34; Age 24)

- A subtype of dynamic (event-oriented) modality.
  - express a tendency of the subject
  - often paraphrasable by 'sometimes' (cf. quantificational modals: Carlson, 1977; Brennan, 1997)

Persisting exceptionality of *can*

Can is known to be **exceptional** in several ways.

- Epistemic only when negative
  (Confirmed here: all 15 epistemic tokens of *can* in YEC are negative)
- Preterite *could* retains transparent past meaning
  (Cf. *would*, which in matrix clauses can only have past habitual meaning)

A new exceptionality: absence of "British *do*" with *can*

British *do*: appearance of *do* below another auxiliary in ellipsis contexts.

Can occurs more often than any other possibility modal in the YEC (Fig 4), yet occurs with *do* only once (2).

(2) If someone wanted to nick your bike they **can do**. (Spkr 57, Age 17)

This is **not** because British *do* requires an epistemic modal.

- British *do* with dynamic *could* in the YEC:

(3) a. You had your own room. Just the basics and bin which you made as friendly as you could do. (Spkr 69; Age 54)
b. But if I wanted to come home in the evening I could do. (Spkr 69; Age 54)

	# of *do*	(# overall)	%
can	1	(1819)	0.051
could	8	(1513)	0.53
may	2	(36)	5.6
might	7	(218)	3.2
total	*18*	*(3727)*	*0.48*

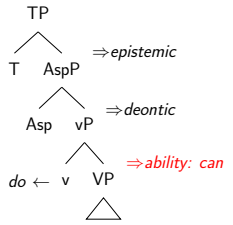
Fig. 4: Rates of British *do* with possibility modals

Discussion

Puzzle: why is *can* exceptional—and why are *could* and *can't* **not**?

Proposal: *can* is associated with a **uniquely low** position, not only below T and Asp, but also below v.

- Suggested in particular by incompatibility with British *do*, if *do* = v stranded by VP ellipsis (Thoms, 2011)



Common structural division:

modals — epistemic = **high** (above A)
root = **low** (below Asp)

(Jackendoff 1972; Zubizarreta 1982; Butler 2003; Hacquard 2006)

Further division proposed here:

root — deontic = **high** (above v)
ability = **low** (below v)

(cf. $\text{Mod}_{\text{Ability/Permission}}$ in Cinque 1999)

Back to *could* and *can't*:

- If *could* and *can't* are just *can* + past tense or negation, why does *could* license *do* and why do both allow epistemic readings?

Further proposal: the **finite tantum** status of modals links *could* and *can't* to higher structural positions.

- Traditional view: *finite tantum* reflects syntactic position (in T).
- Revision: finite tantum reflects fixed *and interpretable* features on modals: *could* bears [PAST], and *can't* bears [NEG]
- *could* and *can't* are consequently syntactically tied to positions outside the in a way that *can* is not.

Conclusions

This study draws on both **variationist** and **formal** perspectives:

- Variationist: quantitative evidence of changes in progress in the modal system, not visible at the level of individual speakers.
- Formal: structural properties explaining *can*'s exceptionality can *also* explain why it resists the general pattern of modal decline.

Next steps: Other varieties of English (Toronto); finite tantum and *spans*; sufficiency readings for *can*; interaction with perfect *have*.

Thanks to Ruth Norman, who worked as a research assistant on this project at Queen's University in Summer 2016.
References and additional data can be found on the paper handout, available online at `http://post.queensu.ca/~bb86/talks/`.

LaTeX TikZposter

	<35	%	36-65	%	>65	%	*Total*	%
Ability	310	*62*	415	*63*	278	*51*	1003	*59*
Circumstantial	155	*49*	178	*44*	128	*31*	458	*41*
Dispositional	37	*90*	32	*65*	17	*57*	86	*72*
Deontic	85	*73*	89	*56*	83	*54*	257	*60*
Epistemic	6	*5*	6	*5*	3	*3*	15	*4*

Figure 5: Instances of *can* by modal interpretations

References

Berglund, Ylva. 1997. Future in present-day English: Corpus-based evidence on the rivalry of expressions. *ICAME journal* 21:7–20.

Brennan, Virginia. 1997. Quantificational modals. *Linguistic Inquiry* 165–169.

Butler, Jonny. 2003. A minimalist treatment of modality. *Lingua* 113:967–996.

Carlson, Gregory. 1977. Reference to kinds in English. Doctoral Dissertation, University of Massachusetts Amherst.

Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. Oxford University Press, USA.

Hacquard, Valentine. 2006. Aspects of modality. Doctoral Dissertation, Massachusetts Institute of Technology.

Jackendoff, Ray S. 1972. *Semantic Interpretation in Generative Grammar.*. Cambridge, MA: MIT Press.

Sankoff, David, Sali Tagliamonte, and Eric Smith. 2005. Goldvarb X: A variable rule application for Macintosh and Windows. *Department of Linguistics, University of Toronto* .

Szmrecsanyi, Benedikt. 2003. *Be going to* versus *will/shall*: Does syntax matter? *Journal of English Linguistics* 31:295–323.

Tagliamonte, Sali. 2002. Comparative sociolinguistics. In *The handbook of language variation and change*, ed. J. K. Chambers, P. Trudgill, and N. Schilling-Estes, 729–63. Malden and Oxford: Blackwell.

Tagliamonte, Sali, and Alexandra D'Arcy. 2007. The modals of obligation/necessity in Canadian perspective. *English World-Wide* 28:47–87.

Tagliamonte, Sali A. 1996-1998. Roots of identity: Variation and grammaticization in contemporary British English. Research grant, Economic and Social Research Council (ESRC), # R00238287.

Tagliamonte, Sali A, and Jennifer Smith. 2006. Layering, competition and a twist of fate: Deontic modality in dialects of English. *Diachronica* 23:341–380.

Thoms, Gary. 2011. From economy to locality: *Do*-support as head movement. Ms., Strathclyde University.

Zubizarreta, Maria Luisa. 1982. On the relationship of the lexicon to syntax. Doctoral Dissertation, Massachusetts Institute of Technology.